TAKOMO

SINCE AI HACKATHON 2026 · CHALLENGE PARTNER PACK

THE FITTER IN YOUR POCKET

Everything your team needs to know about club fitting, Takomo's iron range and how today's online fitting tool decides.

Takomo Golf · Turku, 6–8 November 2026

Includes: this guide + product data files (JSON and CSV)

HOW TO USE THIS PACK

This pack gives you the domain knowledge a club fitter carries in their head, plus a trimmed copy of Takomo's real product data. It explains what the current online fitting tool asks, what it recommends, and which inputs move each recommendation.

- **This guide** explains the fitting outputs, the inputs that drive them, and the key rules the current tool uses.
- **Product data files** (JSON and CSV) list iron heads, shafts and grips with the specs that matter for fitting.

> **New to golf?** Start with Golf basics. Whatever your background, read **What matters most** closely: it lists what every recommendation has to get right.

Treat the rest as a starting point, not a spec. The current rules are deliberately simple. Teams that want an extra challenge will find a bonus on fitting the full bag at the end. You are free to replace, extend or ignore any of them if you can show your approach fits golfers better or explains itself more clearly.

GOLF BASICS FOR FITTING

You don't need to play golf to build a great fitting tool, but a handful of words come up everywhere. This section covers them, and the rest of the pack uses them freely.

THE GAME AND THE GOLFER

- Golfers count strokes over 18 holes, and fewer is better. Par, the score an expert is expected to make, is around 72 on most courses.
- A golfer's typical score shows their skill. Under 75 is a very good player; over 92 is usually a newer or occasional golfer.
- Explanations in this pack assume a right-handed golfer. For a left-handed golfer, left and right are swapped.

THE IRONS

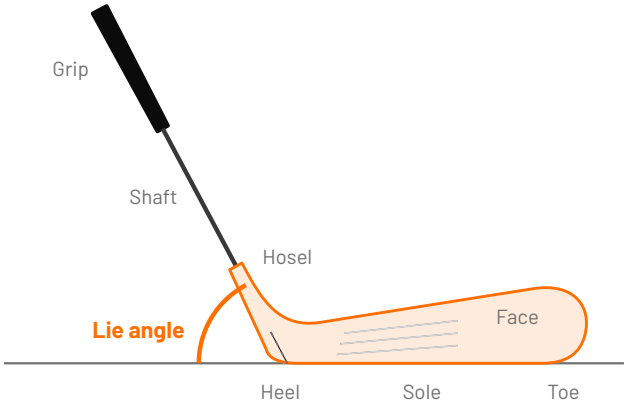
- A set of irons is a series of clubs numbered roughly 4 to 9, plus wedges. Lower numbers have less loft, so the ball flies lower and further.
- The **7-iron** sits in the middle of the set, so fitters use it as the reference club for speed and distance.

PART	WHAT IT IS
Face	The hitting surface.
Sole	The bottom of the head, which rests on the ground.
Toe and heel	The end of the head furthest from the shaft, and the end nearest to it.
Hosel	The neck that joins the head to the shaft.
Loft	How far the face is tilted back. More loft means higher, shorter shots.

PART	WHAT IT IS
Lie angle	The angle between the shaft and the ground when the sole sits flat.
Offset	How far the face sits behind the hosel. More offset gives the golfer slightly more time to square the face.
Cavity back, muscle back	A cavity back is hollowed out behind the face, which spreads weight to the edges and forgives off-centre hits. A muscle back is solid and compact, for control and feel.
Forged	Shaped from a single piece of soft steel. Valued by better players for feel.

PARTS OF AN IRON, AND WHAT LOFT AND LIE MEASURE

Front view



Lie decides whether the toe or heel points up at impact.

Side view, from the toe

Head cut through the middle, ball flies to the right



Less loft
lower, longer

More loft
higher, shorter

Loft = face tilt back from the dashed upright.

Lie is seen from the front, loft and offset from the side. A fitter can change length, lie and shaft on any head; loft and offset come with the head model.

THE SHOT

TERM	WHAT IT MEANS
Swing speed	How fast the clubhead is moving at impact, in mph.
Carry	How far the ball flies in the air. Total distance adds the roll after landing.
Launch	How steeply the ball leaves the face.
Spin	Backspin on the ball. Some keeps the ball in the air and makes it stop on the green; too much costs distance.
Pull, push	A shot that starts left (pull) or right (push) of the target.
Draw, fade	A shot that curves gently left (draw) or right (fade). A hook and a slice are the strong, unwanted versions.
Strike, mishit	Where the ball meets the face. A mishit is off-centre, toward the toe or heel, and loses distance and direction.
Dispersion	How spread out a golfer's shots are. Smaller is better.

TERM	WHAT IT MEANS
Tempo	The rhythm of the swing, from smooth to quick.
Transition	The moment the club changes from going back to coming down. A sharp transition bends the shaft harder.
TrackMan	A radar launch monitor that measures speed, launch, spin, carry and direction for every shot. One is available at the event.

HOW A CLUB FITTING WORKS

An iron fitting chooses six things, and each one changes a different part of the shot. A good fitter collects evidence for each choice separately, then explains them together.

OUTPUT	WHAT IT CHANGES	WHAT A FITTER LOOKS AT
Head model	How forgiving the club is on off-centre hits, how far and high the ball flies, how it feels	Skill level, how consistently the golfer hits the centre of the face, what they value most
Shaft	How high the ball launches, how much it spins, how the club feels during the swing	Swing speed, tempo, desired ball flight, age and strength
Length	Where on the face the ball is hit, and the golfer's posture	Height and arm length
Lie angle	Whether the ball starts left or right of the target	Height and arm length, then the sole marks and start direction of real shots
Grip size	How much the hands turn the club through impact, and comfort	Hand size
Set makeup	Which clubs are in the set, with even distance gaps between them	Distance with each club, other clubs in the bag. Not part of today's tool

Takomo also sells two ways to buy the result:

- **Stock set:** standard specs, faster delivery, 30-day returns.
- **Custom build:** any spec change, built to order, not returnable.

Part of the fitting is therefore deciding whether a golfer really needs custom specs. A small benefit may not be worth losing the returns window.

WHAT MATTERS MOST

Whatever you build, a recommendation is only useful if it gets these six things right.

1. **Swing speed is the most important input.** It sets the shaft flex, helps decide steel or graphite, and steers the head choice. If you can measure only one thing well, measure this.

2. **Skill and priorities choose the head.** The less consistently a golfer hits the centre of the face, the more forgiveness matters. Distance, feel and control trade off against each other, so the golfer's priorities count.

3. **Body measurements set length and lie.** Height and wrist-to-floor are the starting point. How the club actually meets the ground in a real swing is the strongest evidence, if you can capture it.

4. **Every result is a complete spec we can build,** using only the product data:
   - one head;
   - one shaft (model and flex) listed for that head;
   - a length from –1.5″ to +1.5″, in ½″ steps;
   - a lie from 3° flat to 3° upright, in 1° steps;
   - one of three grip sizes;
   - right- or left-handed;
   - stock set or custom build.

5. **Stock or custom is part of the answer.** Any non-standard spec means a build that can't be returned. Recommend it when it clearly helps, and say why.

6. **The golfer should understand why.** Each part of the spec needs a reason in plain words, and a close call should be shown as one.

TODAY'S ONLINE FITTING TOOL

The current tool is a 12-question iron questionnaire. Most answers are ranges (for example 71–80 mph) rather than exact numbers, and each output is then chosen by its own rule. Only irons are fitted; putters have a simple length chart.

#	QUESTION	ANSWER OPTIONS	USED FOR
1	Age	20–70+	Shaft material (70+ gets graphite)
2	Sex	Male / female	Shaft material and flex range
3	Rounds per year	Under 10, 10–30, 31–50, over 50	Not used yet

#	QUESTION	ANSWER OPTIONS	USED FOR
4	Typical 18-hole score	Under 71, 71–77, 78–84, 85–92, over 92	Head model
5	7-iron swing speed **or** 7-iron carry	Speed: under 60, 60–70, 71–80, 81–90, over 90 mph. Carry: under 120, 120–135, 135–155, 155–175, over 175 yd	Head model, shaft flex, shaft material
6	Top 3 desired characteristics	Forgiveness, distance, forged feel, distance control, workability, less offset, more offset	Head model
7	Desired ball flight	Low, medium, high	Shaft model
8	Swing tempo	Smooth, moderate, quick	Shaft model
9	Height	cm or ft/in	Length, lie angle
10	Wrist-to-floor (in golf shoes)	cm or ft/in	Length, lie angle
11	Glove size	XS, S, M, M/L, L, XL	Grip size
12	Handedness	Right / left	Which way the clubs are built

The golfer answers either swing speed or carry, never both, and the five ranges are treated as equivalent. The result screen shows the two best-scoring heads, each with a shaft, length, lie and grip, and a link to either the stock product or the custom builder.

HEAD MODEL SELECTION

Takomo's five iron heads run from most forgiving (101 MKII) to most workable (301 MB). The tool scores every head against the golfer and shows the top two.

HEAD	CONSTRUCTION AND PURPOSE	7-IRON LOFT	OFFSET (7-IRON)	DISTANCE	FORGIVE-NESS	WORK-ABILITY	DISTANCE CONTROL	BEST-FIT SCORE
101 MKII	Forgiveness and distance in a sleek package	29°	3.4 mm	5	5	2	2	Over 92
201 MKII	Blend of distance, forgiveness and workability	31°	2.77 mm	4	4	3	3	85–92
201T MKII	More workable, with a hint of forgiveness	33°	2.46 mm	3	3	4	4	78–84
301 CB	Forged cavity back, control with some forgiveness	34°	1.8 mm	1	2	5	5	71–77
301 MB	Forged muscle back, maximum workability and feel	34°	1.8 mm	1	1	5	5	Under 71

Ratings are Takomo's own 1–5 scale. The 301 heads are forged and rated highest for feel.

WHAT THE TRADE-OFFS MEAN FOR A GOLFER

- **Forgiveness:** a larger head with its weight spread to the edges loses less ball speed and direction on off-centre hits. Most golfers scoring over 85 benefit more from this than from anything else.
- **Distance:** comes mostly from stronger (lower) lofts and faster faces. The 101 MKII's 7-iron is 5° stronger than the 301's, so it flies further partly because it is effectively a different club number.
- **Workability and distance control:** compact heads with less offset let skilled players shape shots and control carry precisely, but punish mishits.
- **Offset:** the face sits slightly behind the hosel, giving the golfer a fraction more time to square the face. More offset helps golfers who slice and launches the ball higher; better players often dislike how it looks.

KEY RULES IN TODAY'S TOOL

Each head gets a total score from three parts, and the two highest totals are shown:

1. **Skill match:** 5 minus how many score buckets the golfer is from the head's best-fit level.
2. **Speed match:** slower swing speeds score the distance-oriented heads higher; faster speeds score the compact heads higher. A high-scoring golfer (over 92) with a fast swing is capped, so speed alone cannot push them into a compact head made for skilled players.
3. **Chosen characteristics:** for each characteristic the golfer picked, the head's rating is added. Forged feel counts double. "More offset" adds the offset rating; "less offset" adds its inverse.

The golfer sees the best match, the runner-up and each head's description and ratings.

SHAFTS

Shaft guidance is the gap customers mention most. Today the tool picks a flex from swing speed, a material from speed, sex and age, and a model from desired ball flight and tempo.

WHAT EACH SHAFT PROPERTY DOES

- **Flex** should match swing speed. Too soft tends to launch higher, spin more and miss left (right-handed). Too stiff tends to fly lower, miss right and feel harsh.
- **Weight** affects tempo and control. Heavier shafts feel more stable and usually fly lower; lighter shafts are easier to swing fast and usually fly higher. Fast or aggressive swingers often lose control with a shaft that is too light.
- **Material:** steel is heavier, more consistent and cheaper; graphite is lighter, absorbs vibration and helps slower swingers launch the ball. Graphite especially suits seniors, many women and golfers with joint pain.
- **Launch and spin profile** comes from where the shaft bends. Each shaft is rated low, mid or high for launch and spin. Actual launch and spin numbers depend on the golfer's speed and delivery, so the profile describes how a shaft behaves relative to the others, not a fixed number.
- **Tempo** matters for feel. Smooth swingers usually prefer lighter shafts that bend more near the head; golfers with a quick transition bend the shaft harder and suit firmer, heavier ones.

SHAFT MODELS AVAILABLE

SHAFT MODEL	MATERIAL	WEIGHT (G)	FLEXES	LAUNCH AND SPIN	TEMPO PROFILE	AVAILABLE ON
KBS Tour	Steel	120–133	R, S, X	Low–mid	Moderate–quick	All heads
KBS Tour Lite	Steel	105–110	R, S, X	Mid–high	Smooth	All heads
KBS C-Taper	Steel	125–140	R, S, X	Low	Quick	301 CB, 301 MB
KBS C-Taper Lite	Steel	109–125	R, S, X	Mid–high	Quick	301 CB, 301 MB
Mitsubishi MMT	Graphite	57–86	L, A, R, S	Low–mid	Moderate–quick	MKII heads
True Temper Denali Silver	Graphite	53–89	L, A, R, S	Mid–high	Smooth	MKII heads
Takomo Obsidian	Graphite	72–111	A, R, S, X	Low–mid	Any	301 CB, 301 MB
Fujikura Atmos Red	Graphite	72	A, R, S	High	Any	301 CB, 301 MB

Flex letters: L = ladies, A = senior, R = regular, S = stiff, X = extra stiff. The data files list each flex as its own shaft with its exact weight.

KEY RULES IN TODAY'S TOOL

7-IRON SWING SPEED	7-IRON CARRY	FLEX
Under 60 mph	Under 120 yd	Ladies (women) or Senior (men)
60–70 mph	120–135 yd	Senior

7-IRON SWING SPEED	7-IRON CARRY	FLEX
71–80 mph	135–155 yd	Regular
81–90 mph	155–175 yd	Stiff
Over 90 mph	Over 175 yd	Extra stiff

- **Material:** graphite if the flex is Ladies or Senior, the golfer is female, or the golfer is 70 or older. Otherwise steel.
- **Model:** within the chosen material and flex, the tool picks the shaft whose flight and tempo profiles both match the golfer's answers. It relaxes the match step by step (flight only, tempo only, nearest flex) when no shaft fits.
- Tempo is used to choose the shaft model; flex comes from swing speed alone.

LENGTH AND LIE ANGLE

Length and lie are both chosen today from two body measurements: height and wrist-to-floor (the distance from the wrist crease to the floor, standing upright in golf shoes, arms hanging). Takomo offers length from –1.5″ to +1.5″ and lie from 3° flat to 3° upright, in ½″ and 1° steps.

WHAT LENGTH DOES

- Too long: the golfer stands more upright or holds lower on the grip, strikes drift toward the heel, and control suffers.
- Too short: the golfer hunches, strikes drift toward the toe, and they may lose some speed.
- Changing length also changes lie in practice: a longer club sits more toe-up at address.

WHAT LIE ANGLE DOES

- Lie is the angle between the shaft and the ground when the sole sits flat.
- **Too upright** (toe up at impact): the face points left of target, so shots start left for a right-handed golfer.
- **Too flat** (toe down at impact): the face points right, so shots start right.
- The effect grows with loft, so it matters more in short irons and wedges than in long irons.
- For left-handed golfers the directions are mirrored.

KEY RULES IN TODAY'S TOOL: LENGTH FROM HEIGHT

HEIGHT (CM)	HEIGHT (FT/IN, APPROX.)	LENGTH
Under 155.5	Under 5′ 1″	–1.5″
155.5–160.4	5′ 1″–5′ 3″	–1″
160.5–168.4	5′ 3″–5′ 6″	–0.5″
168.5–183.4	5′ 6″–6′ 0″	Standard
183.5–191.4	6′ 0″–6′ 3″	+0.5″
191.5–196.4	6′ 3″–6′ 5″	+1″
196.5 and over	6′ 5″ and over	+1.5″

Arm length also counts: for the same height, long arms (a short wrist-to-floor) get a shorter club and short arms a longer one. Ladies-flex shafts are built 1″ shorter, so +1″ is added back for them.

KEY RULES IN TODAY'S TOOL: LIE FROM HEIGHT AND WRIST-TO-FLOOR

Lie uses a classic static fitting chart. For a given height, a longer wrist-to-floor (shorter arms, hands further from the ground) means a more upright lie; a shorter wrist-to-floor (longer arms) means a flatter lie. The result is capped at ±3°.

HEIGHT	WRIST-TO-FLOOR	LIE
152 cm (5′ 0″)	86 cm (34″)	1° upright
173 cm (5′ 8″)	86 cm (33.8″)	Standard
188 cm (6′ 2″)	94 cm (37″)	2° upright
196 cm (6′ 5″)	100 cm (39.5″)	3° upright
193 cm (6′ 4″)	76 cm (30″)	2° flat

A professional fitter treats this static chart only as a starting point. They then check real shots: the mark the sole leaves on a hard board or tape, the shape of the divot (the patch of turf the club takes), and where the ball starts. The online tool has no input like this yet.

GRIP SIZING

Grip size is chosen today from glove size alone. All grips are Lamkin, in three sizes.

GLOVE SIZE	GRIP
XS, S, M	Standard
M/L, L	Midsize
XL	Oversize

WHY SIZE MATTERS

- **Too small:** the fingers dig into the palm, the hands become overactive, and shots tend to pull or hook.
- **Too large:** the hands are restricted through impact, the face stays open, and shots tend to push or fade.
- Comfort and grip pressure matter too. Golfers with arthritis often prefer a larger grip regardless of hand size.

A fitter usually measures hand length (wrist crease to fingertip of the middle finger) and the longest finger, and checks how the fingers of the top hand touch the palm. Many golfers also do not know their glove size, or wear the wrong one.

STOCK VS CUSTOM BUILD

A recommendation can be bought as a stock set only when every spec is standard. Any change at all makes it a custom build, which cannot be returned.

	STOCK SET	CUSTOM BUILD
Shaft	Steel, KBS Tour or KBS Tour Lite	Any available shaft, including all graphite
Length, lie, grip	Standard only	Any available option

	STOCK SET	CUSTOM BUILD
Delivery	Faster	Built to order
Returns	30 days	Not returnable

Key rule in today's tool: the tool links to the stock product if the shaft is a stock steel option and length, lie and grip are all standard. Otherwise it links to the custom builder with the specs pre-filled.

Some recommendations sit just past a standard spec, such as half an inch longer or one degree upright; others need several changes and a graphite shaft. Helping golfers understand how much a custom spec matters for them, and choose between stock and custom with confidence, is part of the challenge.

GETTING GOOD DATA AT HOME

Every recommendation depends on a few numbers the golfer gives us. At home, without a fitter or a launch monitor, those numbers are hard to get right, and this is where the challenge has the most room.

INPUTS THAT ARE HARD TO SELF-REPORT

- **7-iron carry:** golfers tend to remember their best shot rather than their average, and may include roll. Many have never measured it.
- **Lofts differ between brands and models.** Takomo's own 7-irons range from 29° (101 MKII) to 34° (301). Some other brands' 7-irons are 28° or stronger. As a rough rule, 4° of loft is about one club, or 10 yards, of carry, so "my 7-iron goes 150" means different things for different golfers.
- **Swing speed:** few golfers know it.
- **Wrist-to-floor:** hard to measure alone; posture and shoes change it by a few centimetres, which can move the lie by a degree.
- **Glove size:** often guessed or out of date.
- **Typical score:** a recent good round is often reported rather than the average.
- **Tempo and desired ball flight:** newer golfers rarely know either.

HOW TODAY'S TOOL KEEPS IT SIMPLE

The current questionnaire is deliberately short and takes each answer as given. That leaves plenty of space to build on:

- Swing speed and carry are treated as equivalent buckets, and answers are not cross-checked against each other.
- Lie angle comes from body measurements (static fitting). Ball flight, divots and start direction are not part of it yet.
- Grip size comes from glove size only; hand and finger measurements could add detail.
- The same length and lie charts are used for all golfers.
- Rounds per year is collected for future use.
- Results are presented as a single answer, without a confidence level or a personal explanation of why.
- The fitting covers irons. Putters have a simple height chart (160–170 cm: 33″, 171–180 cm: 34″, 181–196 cm: 35″).

PRODUCT DATA FILES

The data comes in one JSON file and three CSVs with the same content. It covers 5 iron heads, 35 shaft options and 3 grips. No prices, customer data or internal identifiers are included.

FILE	CONTENTS
`takomo_products.json`	Everything: heads, shafts, grips, length and lie options, and today's lookup tables (swing buckets, flex, length by height, lie examples, grip by glove size, stock-set rule, putter chart)
`heads.csv`	One row per iron head: loft, offset, ratings, best-fit score level, stock shafts
`shafts.csv`	One row per shaft and flex: material, weight, tempo and flight profile, compatible heads
`grips.csv`	The three grip sizes

FIELD GLOSSARY

FIELD	MEANING
`seven_iron_loft_deg`	Loft of the 7-iron in degrees. Lower = stronger = further.
`seven_iron_offset_mm`	How far the leading edge sits behind the hosel on the 7-iron.
`distance`, `forgiveness`, `workability`, `distance_control`, `feel`, `offset_rating`	Takomo's relative ratings. 1–5, except feel (0–2).
`target_score_label`	Typical 18-hole score of the golfer the head is designed for.
`stock_shafts`	Shaft models available in a returnable stock set for that head.
`build`	Some KBS shafts are listed separately for MKII and 301 heads with slightly different profiles.
`flex`	lady (L), senior (A), regular (R), stiff (S), x-stiff (X).
`weight_g`	Shaft weight in grams.
`tempo_profile`	Swing tempos the shaft suits: slow, mid, fast. Empty or "any" = any tempo.
`flight_profile`	Launch and spin character relative to the other shafts: low, mid, high. Real numbers vary per golfer.
`heads`	Heads the shaft can be fitted to.
`length_options_in`	Length change from standard in inches, −1.5 to +1.5.
`lie_options_deg`	Lie change from standard in degrees. Negative = flat, positive = upright.

The ratings and profiles are expert judgements, not measurements.

QUESTIONS WORTH EXPLORING

These are prompts, not requirements. Pick one, combine several, or ignore them all.

- Which of the questionnaire's inputs could be measured instead of asked, using only a phone, a laptop or a friend?
- How would you tell when a golfer's answers don't fit together, such as a 175-yard 7-iron with a typical score over 92, and what would you ask next?
- How do you compare a golfer's 7-iron distance when their current 7-iron has a different loft from ours?
- What would a fitter say to explain why this shaft, and not the one next to it in the list?
- How do you show that a recommendation is a close call, for example a height right on the border between two lengths?
- When is a custom spec worth giving up the 30-day return, and how would you say that to a golfer?
- What does a newer golfer need to see or hear to trust the result, compared with an experienced one?
- How would you prove that your result matches what a professional fitter would recommend for the same golfer?

BONUS: FITTING THE FULL BAG

Teams that want to go further can extend the fitting from irons to every club in the bag. Treat this as a bonus: a strong iron fitting comes first, and this pack's product data covers irons only.

A golfer may carry up to 14 clubs. A fitter treats them as one system: each club has a job, and together they should cover every distance without gaps or overlaps.

CLUB	ITS JOB IN THE BAG	WHAT A FITTER CHOOSES	WHAT DRIVES THE CHOICE
Driver	The longest shot, hit off a tee	Loft (typically 8–12°), head shape and forgiveness, shaft, length (around 45″)	Driver swing speed, launch and spin, the usual miss (a slice is the most common)
Fairway woods	Long shots from the ground or a tee	Loft (typically 13–21°), shaft	Swing speed, how easily the golfer gets the ball in the air, the gap below the driver
Hybrids	Replace long irons, which many golfers find hard to hit	Loft (typically 17–27°), shaft	Where the iron set should end, and the gap between woods and irons
Irons	Approach shots to the green	Covered in this pack	Covered in this pack
Wedges	Short approaches, and shots around the green and from sand	Loft (typically 46–60°), bounce (how much the sole resists digging into the ground), sole shape	Full-swing distances, the gap below the pitching wedge, how steeply the golfer swings down, ground conditions
Putter	Rolling the ball into the hole on the green	Length, head shape, balance	Height and posture, the shape of the putting stroke

WHAT CARRIES OVER FROM THE IRON FITTING

- **Swing speed:** a golfer's speed scales across the bag. Driver speed is roughly a quarter higher than 7-iron speed for most golfers, so one good measurement informs every club.
- **Tempo and preferred ball flight:** these point toward similar shaft profiles in every club.
- **Tendencies:** a golfer who slices with irons usually slices more with the driver, where low loft makes curve stronger.
- **Body and hands:** height, arm length and grip size apply to every club.

WHAT IS NEW

- **Gapping:** distances between neighbouring clubs should be roughly even, often 10–15 yards. The iron set's lofts decide where wedges and hybrids must fit. Strong-lofted irons like the 101 MKII leave a bigger gap below the pitching wedge, which may call for an extra wedge.
- **Driver launch and spin:** small changes in loft and shaft have a large effect on driver distance, and they are best judged from real ball flight.

- **Wedges and putters** depend more on technique and course conditions than on speed.

A full-bag fitting should still meet every point in What matters most: complete, buildable specs, a clear reason for each club, and honest close calls.